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Instant Payments Namibia

PERSON-TO-PERSON PAYMENTS SOLUTION

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1. Repository URL

https://github.com/inkandjabanga/IPN

2. ReadMe Documents

A. **Web application documentation**

https://github.com/inkandjabanga/IPN/blob/master/README.md

B. Mobile App Documentation

https://github.com/inkandjabanga/IPN/blob/master/IPN_Mobile/README.md

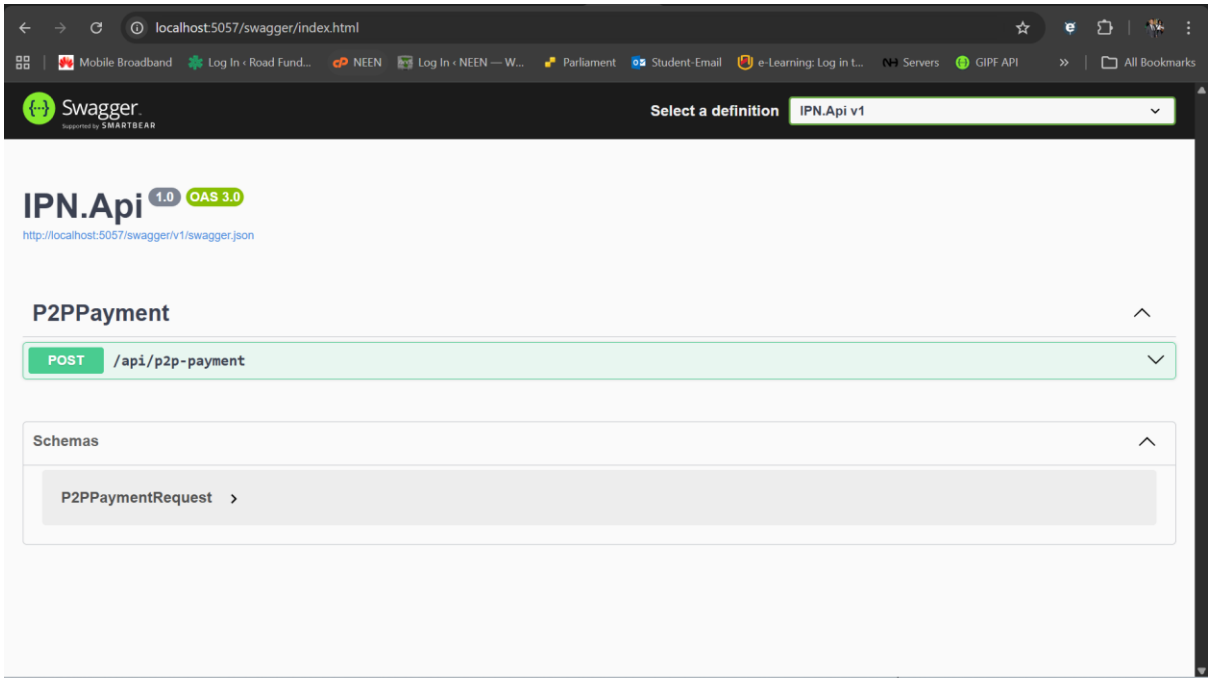
3. Application Screens

The application is built for both mobile and web users. It's ideal that every solution has two versions to make it convenient for the users. Therefore, I built two platforms, a web version and a mobile version of the solution.

3.1. Application Programming interface (API)

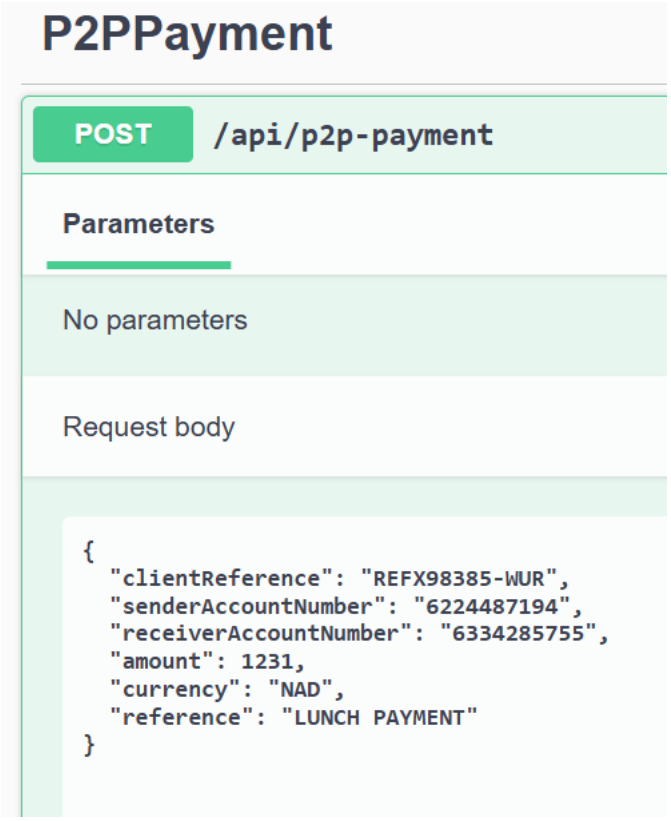
The INP API is documented with the Swagger tool and its accessible from this url on your local machine:
http://localhost:5057/swagger/index.html

- This is a .Net core (C#) Restful API
- The API consists of one Post endpoint: **/api/p2p-payment**



3.1.1. Testing the endpoint

a) SUCCESSFUL PAYMENT



SUCCESSFUL API RESPONSE

Request URL

```
http://localhost:5057/api/p2p-payment
```

Server response

Code	Details
200	Response body


```
{
  "status": "SUCCESS",
  "errorCode": null,
  "transactionId": "TXN20260313002554",
  "message": "Payment processed successfully"
}
```


b) DUPLICATE PAYMENT CLIENT REFERENCE

P2PPayment

POST /api/p2p-payment

Parameters

No parameters

Request body


```
{
  "clientReference": "REFX98385-WUR",
  "senderAccountNumber": "6224487194",
  "receiverAccountNumber": "6334285755",
  "amount": 1231,
  "currency": "NAD",
  "reference": "LUNCH PAYMENT"
}
```


API RESPONSE

http://localhost:5057/api/p2p-payment

Server response

Code	Details
400 *Undocumented*	Error: Bad Request **Response body**


```
{
  "status": "FAILED",
  "errorCode": "ERR001",
  "transactionId": null,
  "message": "Duplicate client reference"
}
```


c) **VALIDATING SENDER AND RECEIVER ACCOUNT NUMBERS**

P2PPayment

POST /api/p2p-payment

Parameters

No parameters

Request body


```
{
  "clientReference": "REF-ORITU8484754",
  "senderAccountNumber": "1234567890ERER",
  "receiverAccountNumber": "1234512313ERTER",
  "amount": 1250,
  "currency": "NAD",
  "reference": "LUNCH PAYMENT"
}
```

Request URL

```
http://localhost:5057/api/p2p-payment
```

Server response

Code	Details
400 *Undocumented*	Error: Bad Request Response body


```
{
  "status": "FAILED",
  "errorCode": "ERR002",
  "transactionId": null,
  "message": "Invalid account number format"
}
```


d) CURRENCY VALIDATION

API REQUEST BODY

Request body


```
{
  "clientReference": "REF-ORITU8484754",
  "senderAccountNumber": "1234567890",
  "receiverAccountNumber": "1234512313",
  "amount": 10,
  "currency": "",
  "reference": "LUNCH PAYMENT"
}
```


API RESPONSE BODY

Request URL

```
http://localhost:5057/api/p2p-payment
```

Server response

Code	Details
400 *Undocumented*	Error: Bad Request Response body


```
{
  "status": "FAILED",
  "errorCode": "ERR003",
  "transactionId": null,
  "message": "Invalid currency"
}
```


e) **INSUFFIENT FUNDS VALIDATION**

POST /api/p2p-payment

Parameters

No parameters

Request body


```
{
  "clientReference": "REF-ORITU8484754",
  "senderAccountNumber": "1234567890",
  "receiverAccountNumber": "1234512313",
  "amount": 100000,
  "currency": "NAD",
  "reference": "LUNCH PAYMENT"
}
```


API RESPONSE

Request URL

```
http://localhost:5057/api/p2p-payment
```

Server response

Code	Details
402 *Undocumented*	Error: Payment Required **Response body**

```
{
  "status": "FAILED",
  "errorCode": "ERR005",
  "transactionId": null,
  "message": "Insufficient funds"
}
```

f) REQUIRED FIELDS VALIDATION

POST /api/p2p-payment

Parameters

No parameters

Request body

```
{
  "clientReference": "",
  "senderAccountNumber": "1234567890",
  "receiverAccountNumber": "1234512313",
  "amount": 1231,
  "currency": "NAD",
  "reference": "LUNCH PAYMENT"
}
```

API RESPONSE

Request URL

```
http://localhost:5057/api/p2p-payment
```

Server response

Code	Details
400 *Undocumented*	Error: Bad Request Response body

```
{
  "status": "FAILED",
  "errorCode": "ERR001",
  "transactionId": null,
  "message": "Missing required field: clientReference"
}
```

Invalid amount validation

P2PPayment

POST /api/p2p-payment

Parameters

No parameters

Request body

```
{
  "clientReference": "4564",
  "senderAccountNumber": "234545745745",
  "receiverAccountNumber": "4574574574574",
  "amount": 0,
  "currency": "NAD",
  "reference": "Lunch payment"
}
```

API RESPONSE

Request URL

```
http://localhost:5057/api/p2p-payment
```

Server response

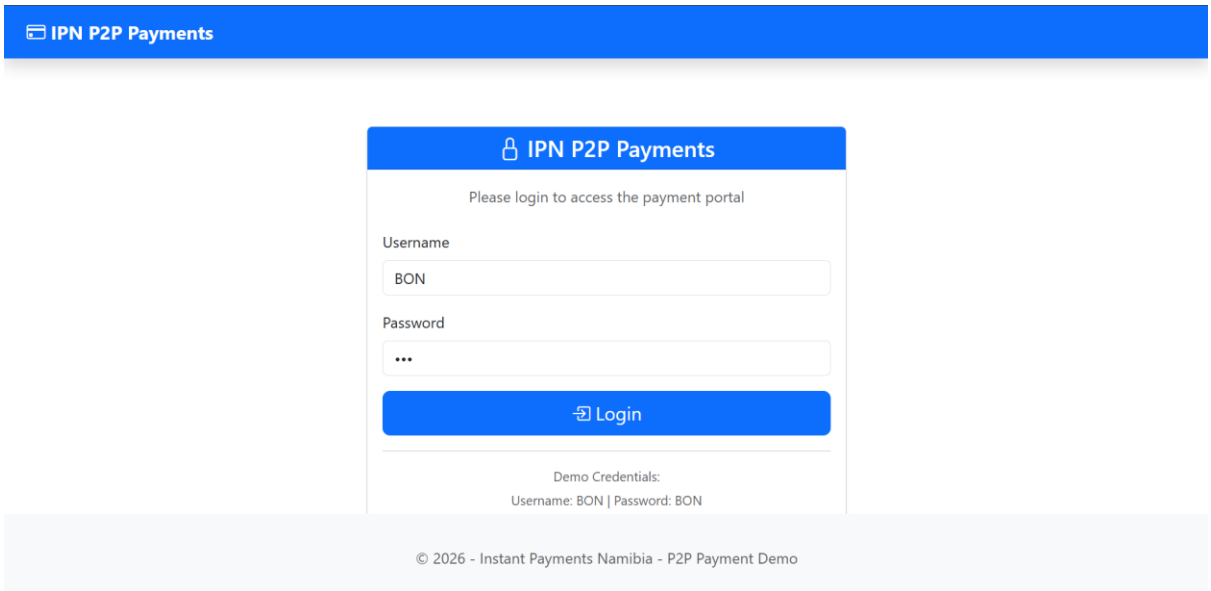
Code	Details
400 *Undocumented*	Error: Bad Request **Response body**

```
{
  "status": "FAILED",
  "errorCode": "ERR004",
  "transactionId": null,
  "message": "Invalid amount"
}
```

3.2. Web Application Screens

3.2.1. Accessing the application

- For security reasons, an optional login screen was added to the application
- A default username and password are pre-populated in the username and password fields (username: **BON** password: **BON**)



➢ Upon a successful login, a user is redirected to the mini dashboard and payment page.

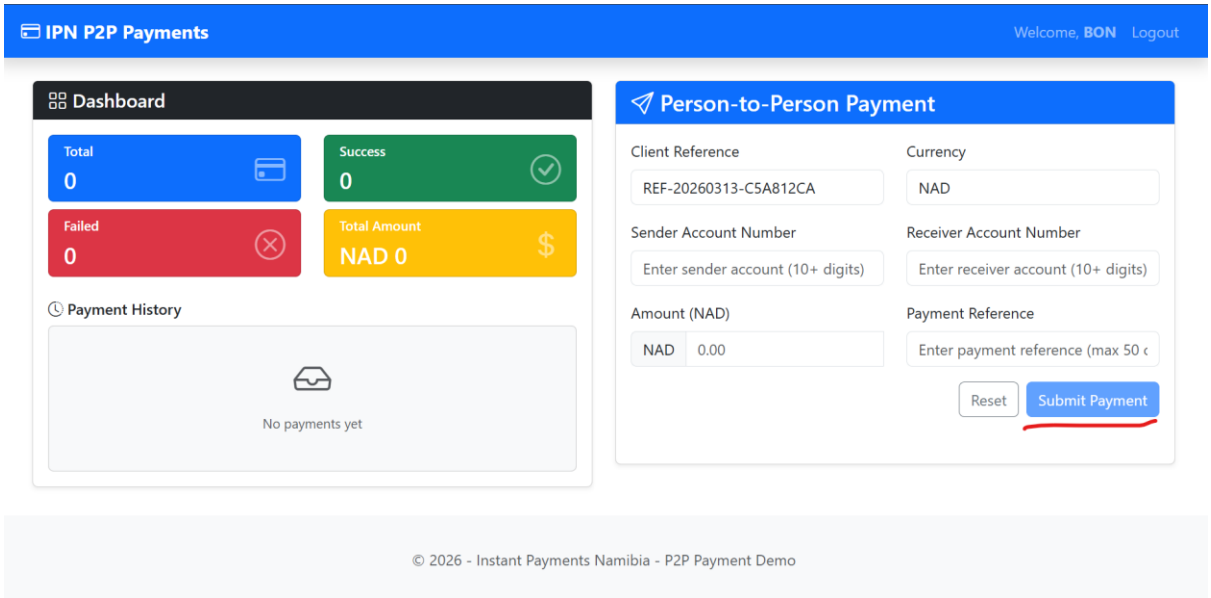
The payment form

- The Person-to-Person payment form is displayed

- User can now enter the payment information
- The submit button is displayed until all payment information are valid. It activates automatically once all fields are valid.
- The client reference number is auto generated by the client application (system)
- The currency is set to **NAD** to avoid user mistakes and its ready only

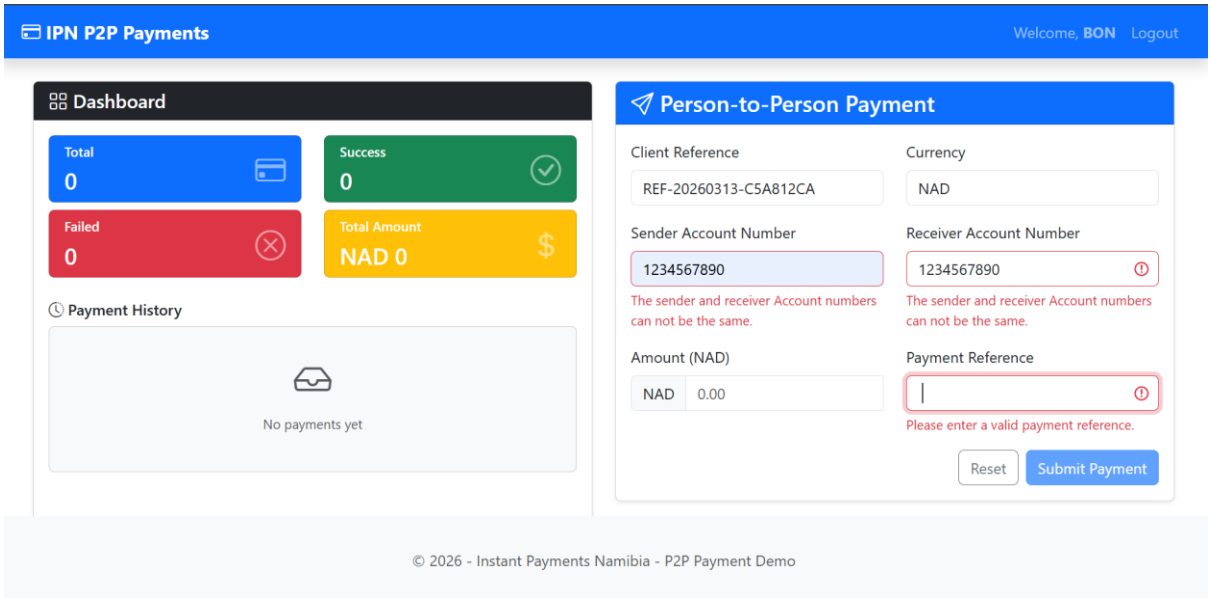
The mini-Dashboard

- The user is presented with few statistics of his/her payment history
- Total number of payments paid (both successful and failed)
- A number of successful payments
- A number of failed payments
- And overall total amount paid

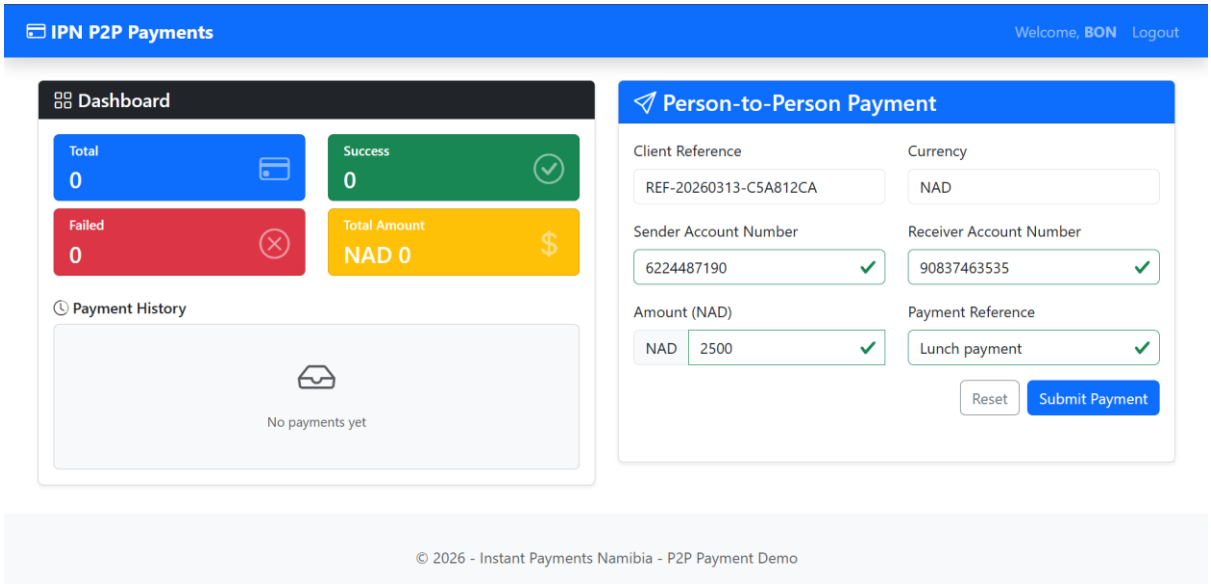


A. FORM VALIDATIONS

- The submit payment button remains inactive until all fields are valid

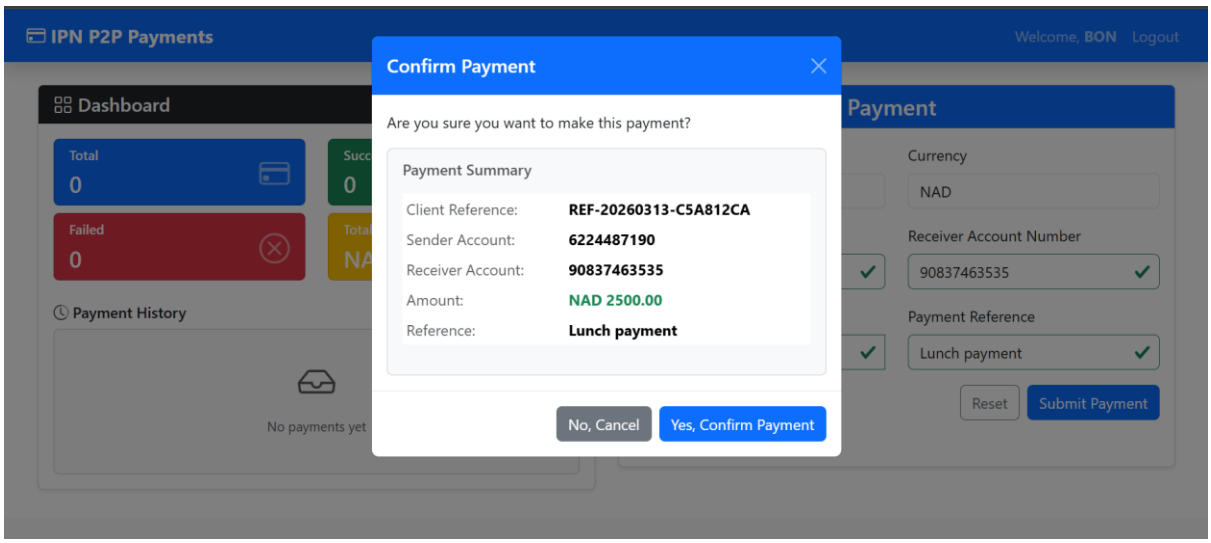


- All fields are valid and the submit payment button is activated



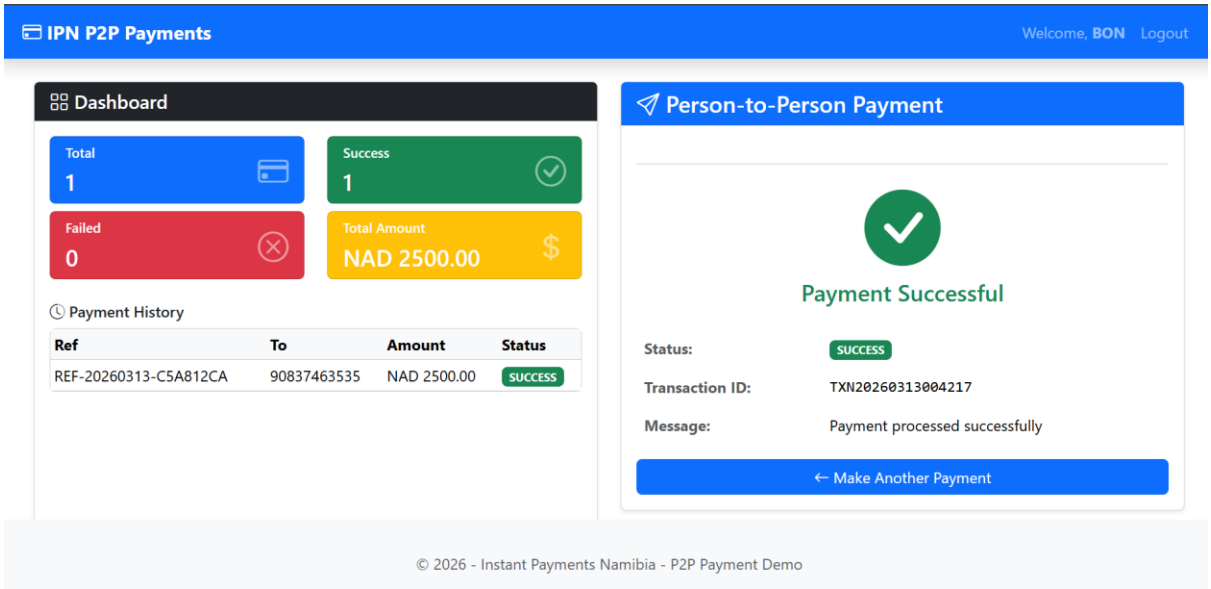
B. Payment Confirmation

- A user Is prompted to confirm the payment



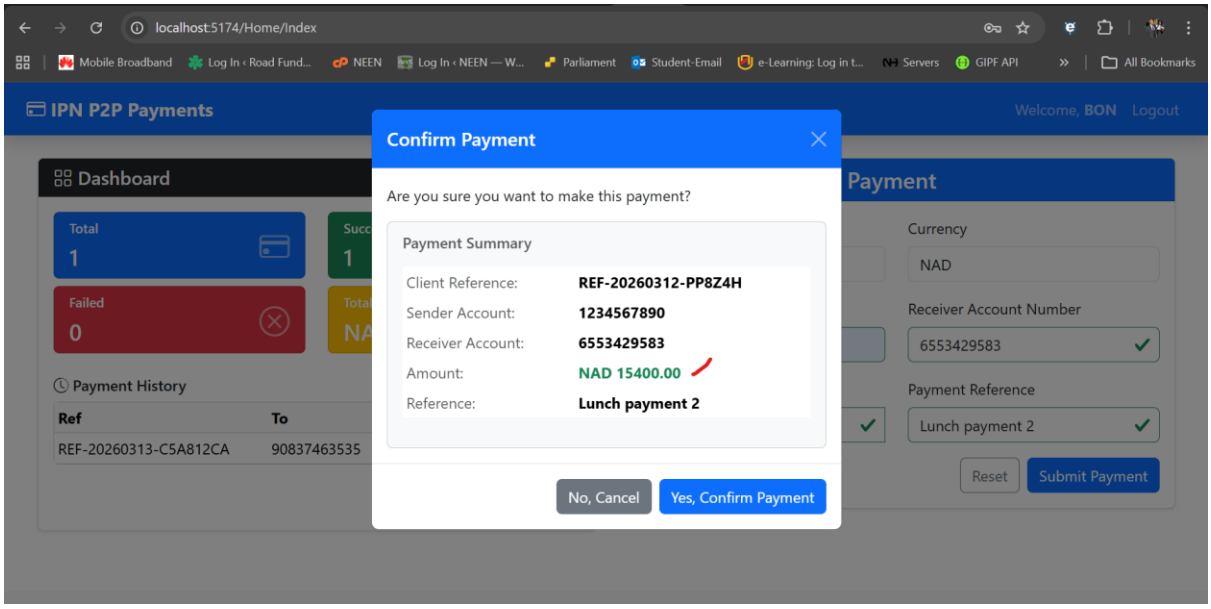
C. Successful Payment

- Upon a successful payment, the dashboard is updated, and the user is notified
- The is given an option to submit another payment request

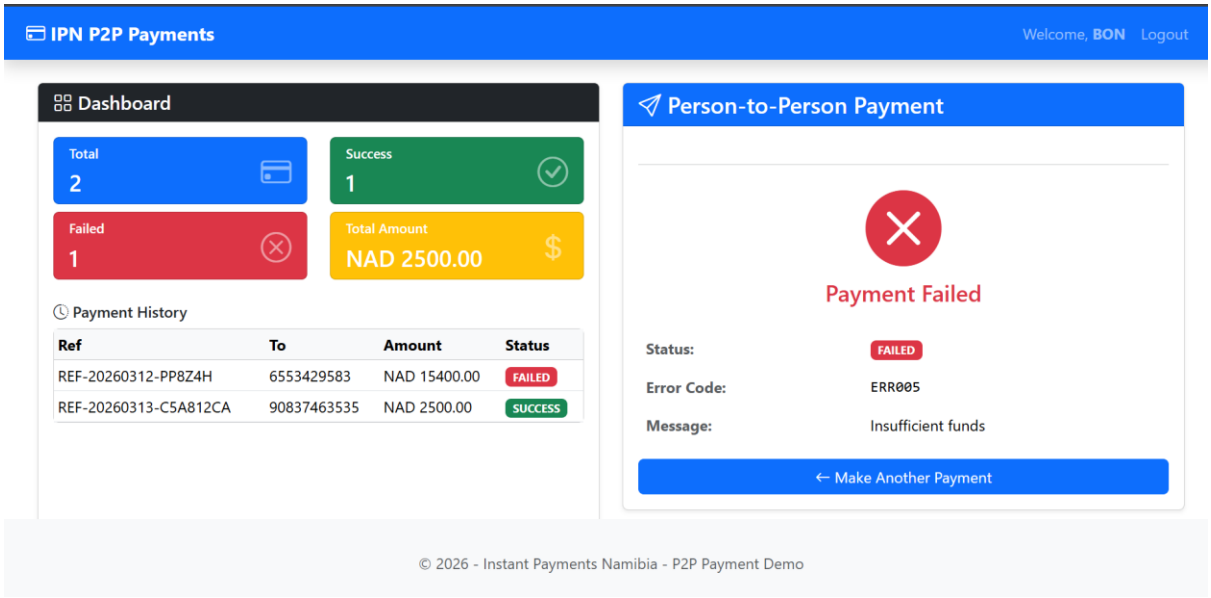


D. Submitting another payment request

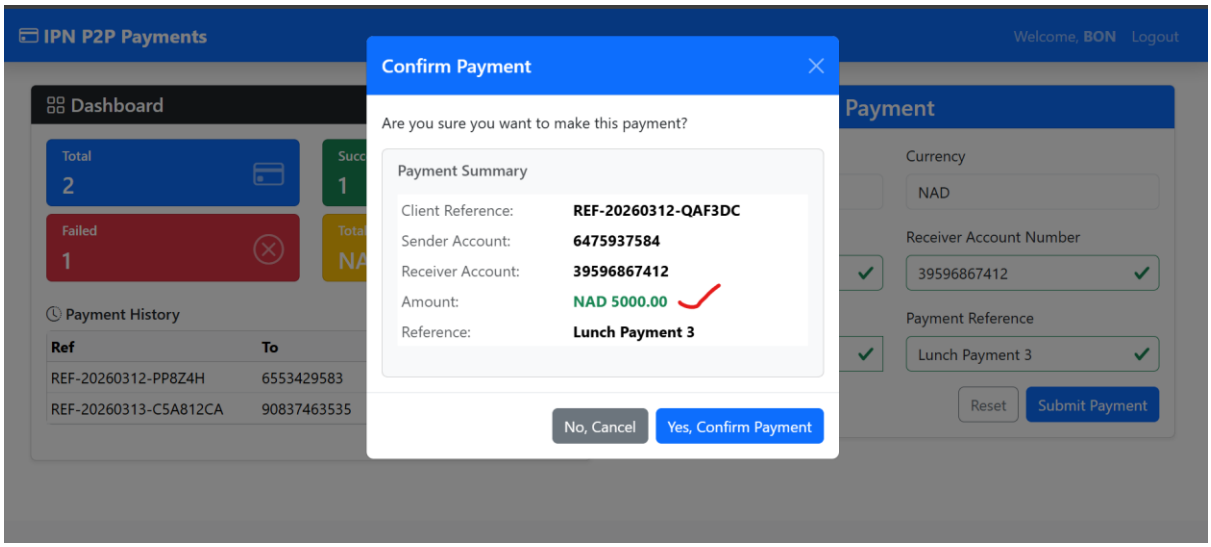
- Insufficient funds on the account
- Payment fails



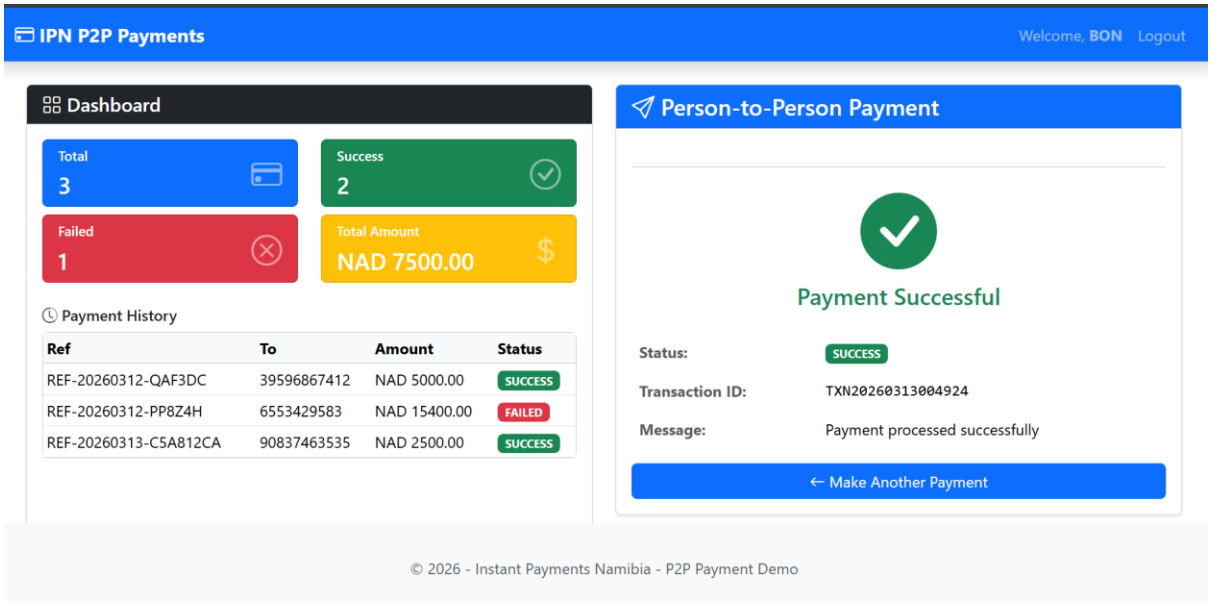
- Payment Fails due to insufficient funds
- The user only has 9999.00 NAD in the account



E. Submitting another successful payment to update the dashboard.



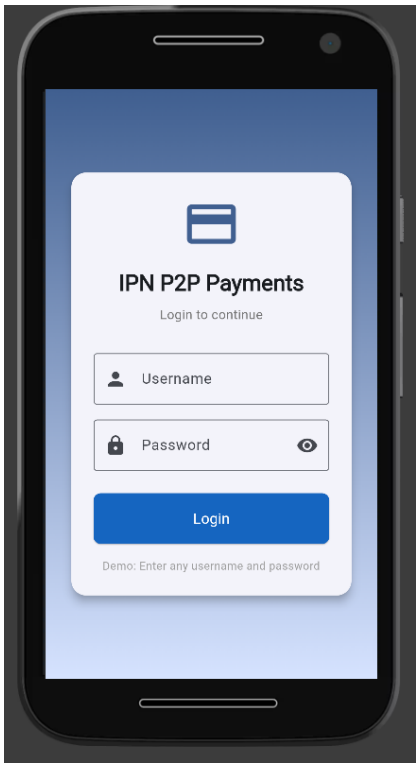
- The Dashboard is updated with numbers of payments made, successful payments, failed payments and total amount paid.



3.3. Mobile App Screens

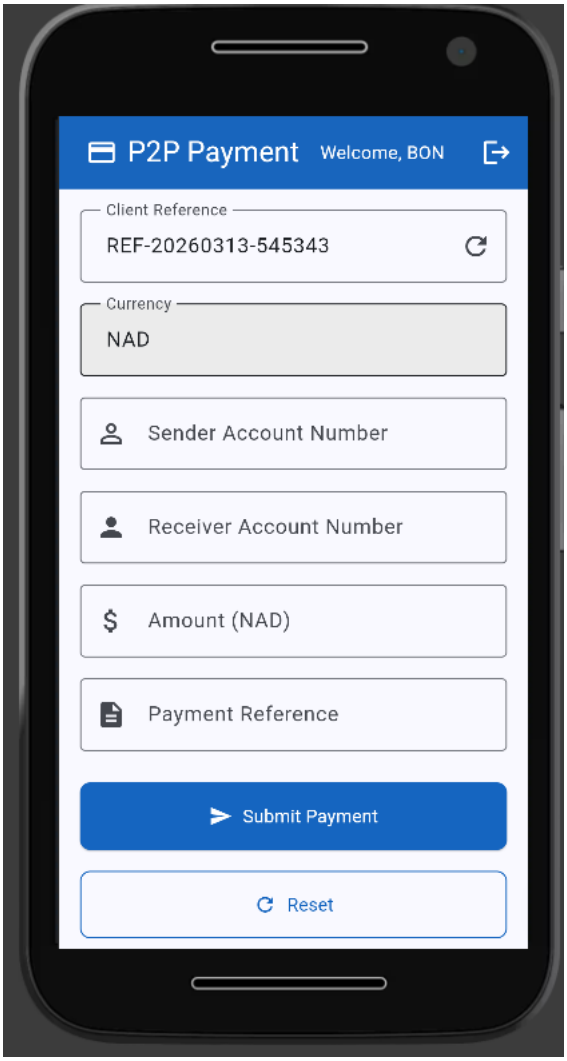
A. LOGIN SCREEN

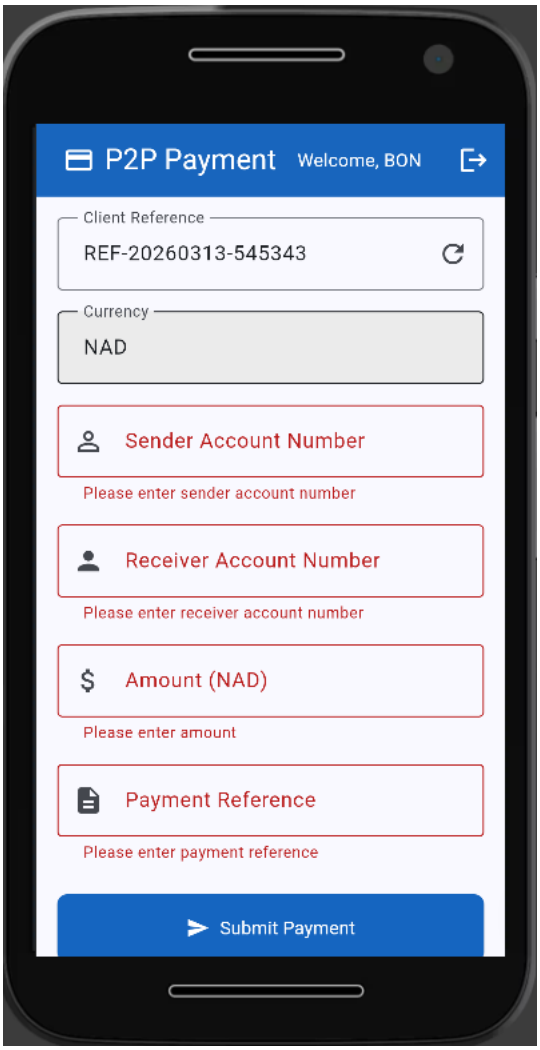
- Enter any username and password (This is simulated)
- Click login



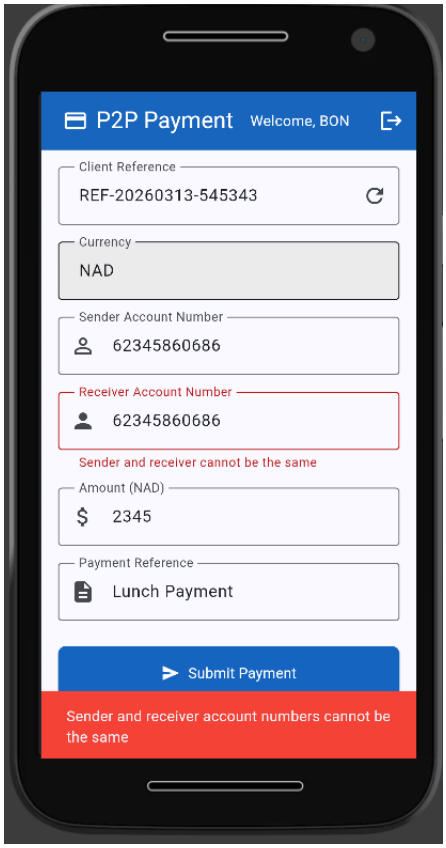
B. Fill in the payment details on the payment screen

- Validation of the fields

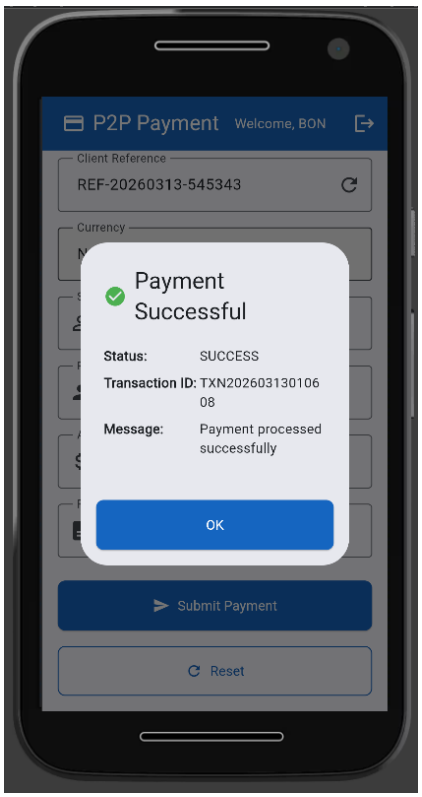




C. Validating the sender and receiver accounts

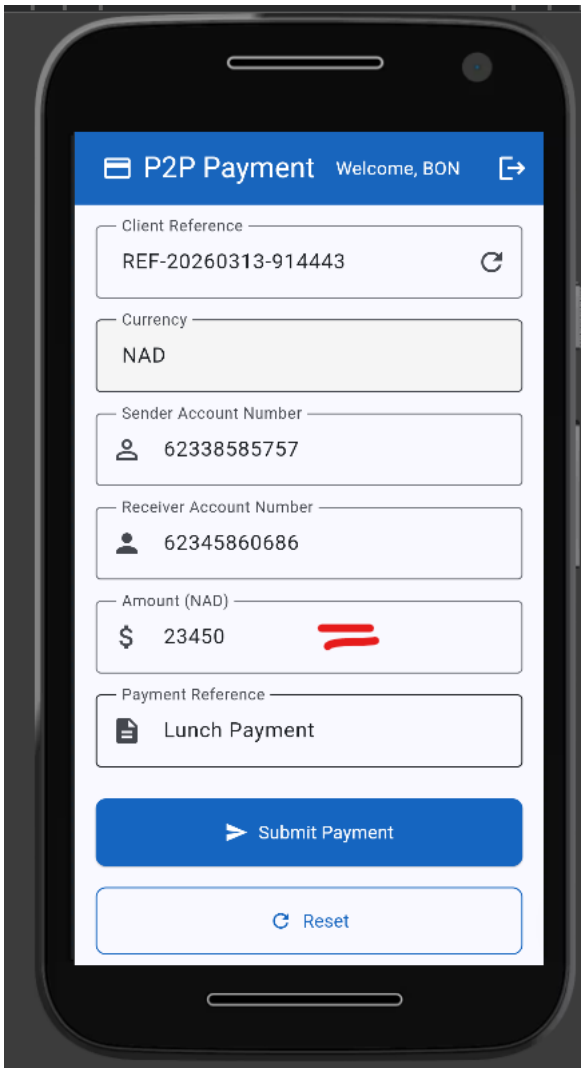


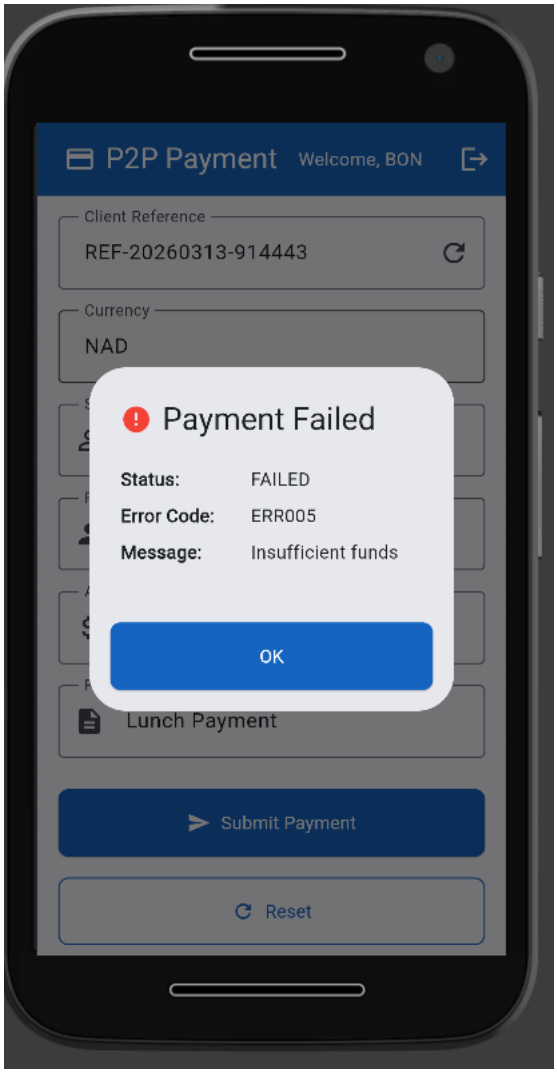
D. Successful Payment Request



E. Failed payment Request

- Insufficient funds




P2P Payment
Welcome, BON
Client Reference
REF-20260313-914443
Currency
NAD
Payment Failed
Status: FAILED
Error Code: ERR005
Message: Insufficient funds
OK
Lunch Payment
Submit Payment
Reset